CityTech TTP Summer Bootcamp

The Command Line

2026-06-03

Agenda

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3. Merge Conflicts
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5. Share your repos
6. Why the Command Line
7. Navigating and Files
8. Getting Help
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Review

What are the most important things you remember from yesterday?

Integrating Changes

Three ways git brings one branch's work into another:

- **Fast-forward** — `main` hasn't moved, so its pointer just slides forward. No new commit.
- **Merge** — histories diverged; git makes a **merge commit** that joins them.
- **Rebase** — replays your commits on top of the latest `main` for a **linear** history.

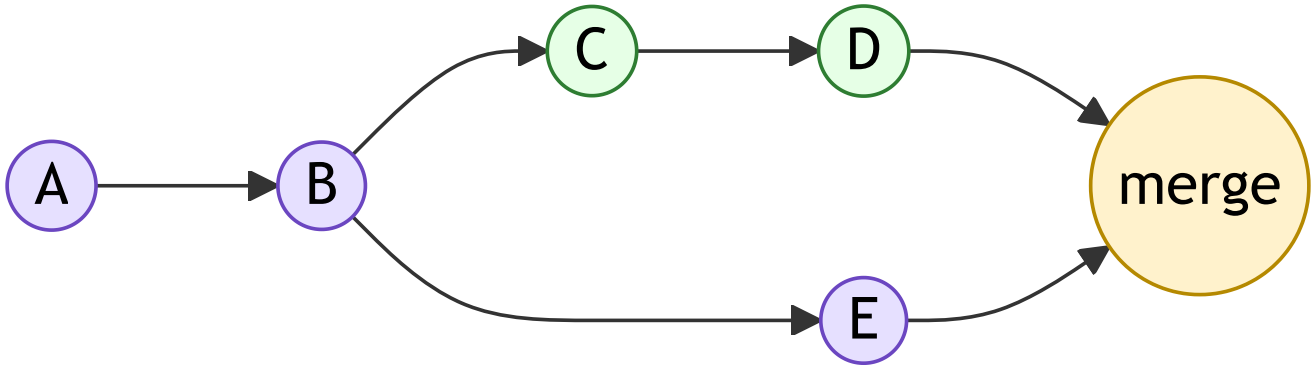
Fast-forward

If `main` has no new commits since you branched, git simply moves the `main` pointer forward to your branch's tip — no merge commit needed.

Green = your branch's commits.

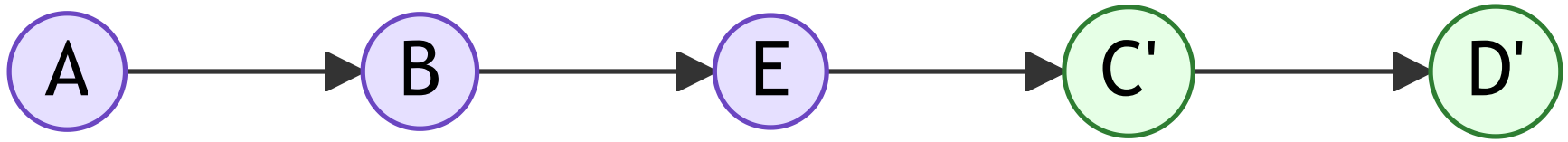
Merge

When **both** branches have new commits, git ties them together with a dedicated **merge commit** (yellow). History keeps both lines visible.



Rebase

Rebase **replays** your branch's commits on top of the latest `main`, rewriting them (`C → C'`) so history stays **linear** — as if you'd started from the newest code.



The 3 Strategies, ELI5

Everyone's writing in one shared notebook:

- **Fast-forward** — no one added pages while you were gone, so yours go right on the end.
- **Merge** — two people wrote at once, so we tape both sets of pages in and add a sticky note: "joined here."
- **Rebase** — you neatly recopy your pages to sit *after* everyone else's newest pages — one clean stack, no sticky note.

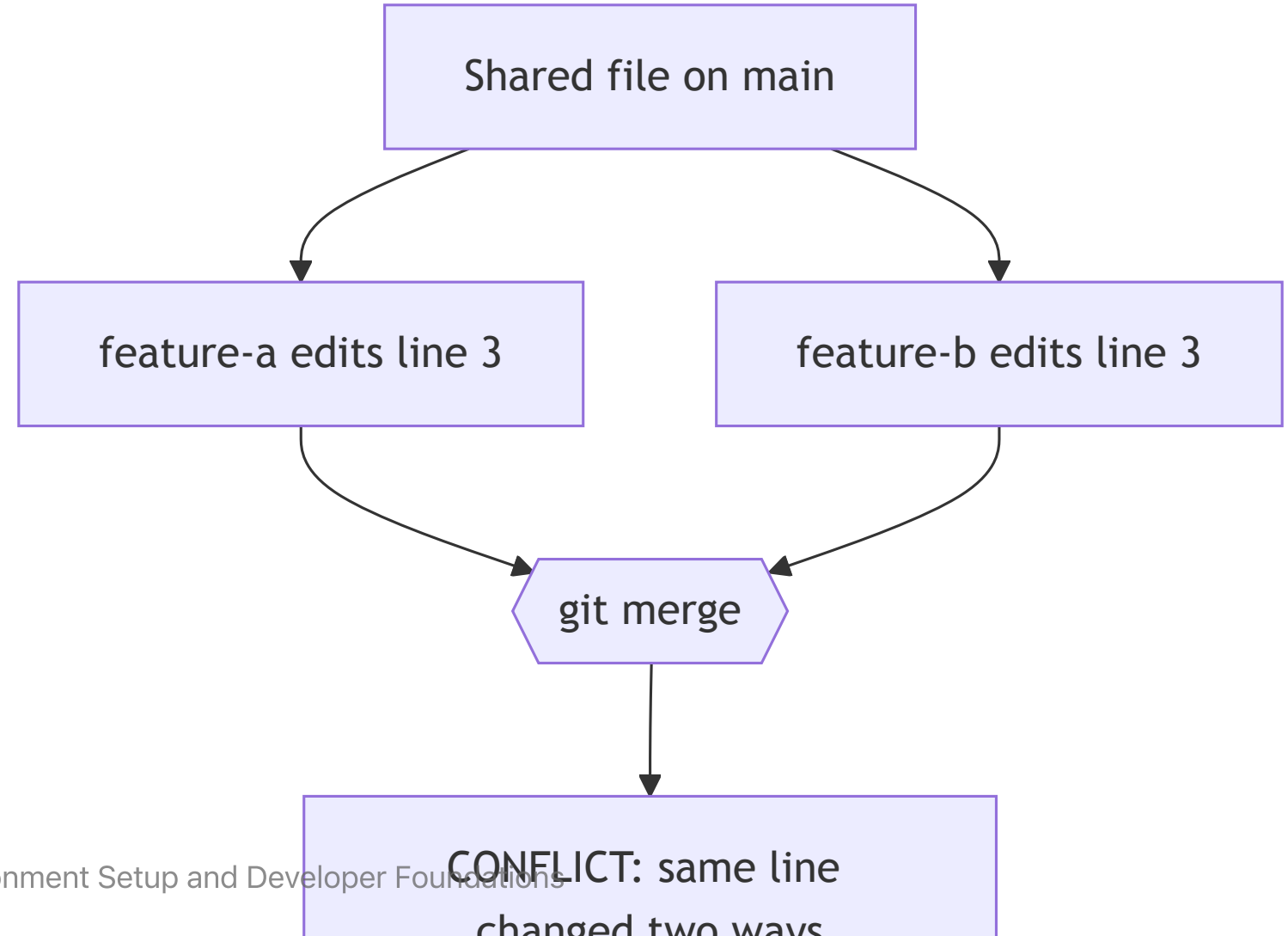
Merge Conflicts

A **merge conflict** happens when two branches change the **same lines** of a file in different ways, and git can't decide which version to keep.

- Common when you and a teammate edit the same file
- git pauses the merge and asks **you** to choose
- They're normal — not a sign you did something wrong

> **No conflict if** you edited different files — or if your changes touch different parts of a file that git can merge automatically.

Why a Conflict Happens



What a Conflict Looks Like

git marks the clashing section right in the file:

```
<<<<<<< HEAD
greeting = "Hi"
=======
greeting = "Hey"
>>>>>>> feature-b
```

- `<<<<<<< HEAD` — your current branch's version
- `=======` — the divider
- `>>>>>>> feature-b` — the incoming branch's version

Resolving a Conflict — Edit It

The most common way: open the file in **VS Code** (or any editor) and choose what to keep.

- Delete the `<<<<<<<` / `=======` / `>>>>>>>` markers, leaving the version you want
- VS Code adds *Accept Current / Incoming / Both* buttons
- …and a 3-way **Merge Editor** for side-by-side resolution

Resolving a Conflict — Take One Side

When you just want one branch's version of the whole file, pick it from the CLI:

```
git checkout --ours   greeting.js   # keep your branch's version
git checkout --theirs greeting.js   # keep the incoming version
```

Resolving a Conflict — Finish or Back Out

Once the file looks right, mark it resolved:

```
git add greeting.js
git commit
```

Or throw away the merge and start over:

```
git merge --abort
```

> `git status` lists every file still in conflict.

Continue 21 Questions Game

https://github.com/jonathan-chin/citytech-ttpr-2026-summer-github-practice-21-questions

Share your repos!

https://forms.gle/vrK7GPxsEPmZ2YvV8

The Command Line

Talking to your computer with text

Why the Command Line?

- It's fast and scriptable once it's in your fingers
- A **GUI** (graphical user interface) is resource-intensive — it needs more powerful, more expensive hardware
- Not every task needs that — a plain terminal runs fine on cheap, minimal machines
- Remote machines (servers) often have no graphical interface at all
- Windows users: do all of this in your **WSL (Ubuntu)** terminal

Anatomy of a Command

A command is usually: `command [options] [arguments]`

```
ls -l /home
# command: ls    option: -l    argument: /home
```

Where Am I? — `pwd`, `ls`

```
pwd             # print working directory (where you are)
ls              # list files here
ls -a           # include hidden files (dotfiles)
ls -l           # long format: permissions, size, date
ls -la          # both at once
```

Moving Around — `cd`

```
cd projects          # into the "projects" folder
cd ..                # up one level
cd ~                 # your home directory
cd /                 # the root of the filesystem
cd -                 # back to the previous directory
```

Paths: `~` = home, `.` = here, `..` = parent.

Absolute paths start at `/`; **relative** paths start from where you are.

"Directory" vs. "Folder"

- They mean the **same thing**: a container that holds files and other directories.
- **Directory** is the older, command-line term — it's why we have `cd` (**c**hange **d**irectory) and `mkdir` (**m**a**k**e **dir**ectory).
- **Folder** is the GUI term, from the file-explorer "folder" icon metaphor.
- Use them interchangeably — on the command line you'll mostly hear **directory**.

Making Things — `mkdir`, `touch`

```
mkdir week1                    # make a directory
mkdir -p week1/exercises       # make nested directories
touch notes.md                 # create an empty file (or update its timestamp)
```

Copying and Moving — `cp`, `mv`

Both follow the same shape:

```
cp source destination     # copy
mv source destination     # move or rename
```

Examples:

```
cp notes.md notes-backup.md     # copy a file
mv notes.md week1/              # move into a folder
mv old.md new.md                # rename
```

Deleting — `rm` (careful!)

```
rm notes.md           # delete a file
rm -r week1-copy      # delete a directory and everything in it
```

⚠️ There is **no Trash or Recycle Bin** on the command line.

Deletes are permanent — read the line twice before you press Enter.

Looking Inside Files

```
cat notes.md          # print the whole file
less notes.md         # scroll a long file (press q to quit)
head notes.md         # first 10 lines
tail notes.md         # last 10 lines
tail -f server.log    # follow a file as it grows
```

Getting Help and Moving Faster

```
man ls            # manual page (q to quit)
ls --help         # quick help for most commands
history           # commands you've run
clear             # clean up the screen
```

- **Tab** completes file and command names
- **↑ / ↓** scroll through past commands
- **Ctrl + C** cancels the running command

When you're stuck:

- Ask other engineers (peers, seniors)
- Read The Manual (**RTM**)
- Ask AI

Play to Practice

1. https://cmdchallenge.com/
2. https://overthewire.org/wargames/bandit/
3. https://web.mit.edu/mprat/Public/web/Terminus/Web/main.html
4. https://github.com/veltman/clmystery